

Area of Triangles

Flagship

Lesson 5-3-flagship

Name: _____

Date: _____

Class: _____

SKY HARBOR MISSION

| The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*If the bottom of a triangle is 10 cm, then $b = 10$ cm
in the formula $A = \frac{1}{2} \times b \times h$*

Base

A side of the triangle you use to find the area.

*A dashed vertical line from the top point straight
down to the base at a 90° angle*

Height

The straight-up distance from the base to the top corner.

A triangle with $b = 8$ and $h = 6$ has area $= \frac{1}{2} \times 8 \times 6 = 24$ sq units

Area

How much space is inside a flat shape.

*The corner of a book or a door frame meets at
exactly 90°*

Perpendicular

Two lines that meet to make a square corner (90 degrees).

*A house shape = a rectangle (walls) + a triangle
(roof)*

Composite figure

A shape made by putting two or more simple shapes together.

*$A = \frac{1}{2} \times b \times h$ means Area equals one-half times
base times height*

Formula

A math rule written with symbols.

Key Ideas & Notes

- Your architecture firm is designing a park with triangular garden beds.
- The client wants to know exactly how much soil to order, which means calculating the area of each triangular section.
- The first garden has a base of 12 feet and a height of 8 feet.
- Plot the vertices of a triangle with base 12 and height 8.

Think About It

- What shape are the garden beds?
- What measurements do we need to find the area?
- How is the area of a triangle related to the area of a rectangle?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

At the build site you survey a triangular garden bed with a base of 12 feet and a height of 8 feet. Which measurement is the height, and how is it different from the slanted side?

Level 1 support

Start here: The height is the perpendicular distance from the base to the top corner.

The height is the ____ distance from the base to the top corner.

La altura es la distancia ____ desde la base hasta la esquina de arriba.

It is different from the slanted side because ____.

Es diferente del lado inclinado porque ____.

Word bank: **base** **height** **perpendicular** **area** **formula**

Listen for: Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Level 2

Does it matter which side of the triangular panel you call the base? Justify your thinking using the formula $A = 1/2 \times b \times h$.

- It ____ matter which side is the base because ____.
- No matter which base I choose, the area is the same since ____.

EXPLORE

Your triangular panel (base 12, height 8) fits inside a 12 by 8 rectangle. Why do we divide by 2 to find the triangle's area before cutting the glass?

Level 1 support

Start here: A triangle is exactly half of a rectangle with the same base and height.

A triangle is half of a ____ with the same base and height.

Un triángulo es la mitad de un ____ con la misma base y altura.

So the area is $\frac{1}{2}$ times 12 times 8, which equals ____.

Entonces el área es $\frac{1}{2}$ por 12 por 8, que es ____.

The rectangle's area is ____, so the triangle is ____.

El área del rectángulo es ____, entonces el triángulo es ____.

Word bank: (base) (height) (area) (perpendicular) (formula)

Listen for: A strong answer states the triangle is half of the $12 \times 8 = 96$ rectangle, so the panel area is 48 sq ft, and connects the $\frac{1}{2}$ to that halving.

Level 2

Two glass panels both have area 48 sq ft. Must they have the same base and height?

Justify or give a counterexample.

- They ____ have the same base and height because ____.
- A counterexample is a triangle with base ____ and height ____, which also gives ____.

CONNECT

At the client review, the homeowner paints a triangular A-frame wall with base 20 ft and height 14 ft; one can covers 50 sq ft. Talk through how many cans the budget needs.

Level 1 support

Start here: Find the triangle area first, then divide by what one can covers.

The wall's area is $1/2$ times 20 times 14, which equals ____ square feet.

El área de la pared es $1/2$ por 20 por 14, que es ____ pies cuadrados.

She needs ____ cans because ____ divided by 50 equals ____.

Necesita ____ latas porque ____ entre 50 es ____.

Word bank: triangle base height area formula

Listen for: Students compute $\text{area} = 140 \text{ sq ft}$, then $140 / 50 = 2.8$, and reason she must round up to 3 cans to cover the whole wall.

Level 2

Why can't the budget order exactly 2.8 cans? Explain what rounding up means in this real situation and how it changes the leftover paint.

- She must round up to ____ cans because ____.
- The leftover paint covers about ____ sq ft, since ____.

Guided Examples

Example 1

What is the area of a triangle with base 10 cm and height 6 cm?

Solution: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = 30$ square centimeters.

Answer: A. 30 sq cm

Example 2

A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

Solution: $24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6$ ft.

Answer: A. 6 ft

Example 3

What is the area of a triangle with base 14 ft and height 6 ft?

Solution: $A = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42$ sq ft.

Answer: A. 42 sq ft

Write About the Math

The Writing Revolution

I can explain my steps using the words base, height, area, and perpendicular.

1. Kernel Sentence subject + verb

Model: Area is how much space is inside a flat shape.

Área es cuánto espacio hay dentro de una figura plana.

Write a kernel sentence about area. Use a subject and a verb.

Escribe una oración base sobre área. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Area matters in math

Área importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Area matters in math because ____.

Área importa en matemáticas porque ____.

but
pero

Area matters in math, but ____.

Área importa en matemáticas, pero ____.

so
entonces

Area matters in math, so ____.

Área importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about area.
Di un hecho verdadero sobre area.

Area ____.

Question *Pregunta*

Ask a question about area.
Haz una pregunta sobre area.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about area.
Muestra entusiasmo sobre area.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with area.
Dile a un compañero qué hacer con area.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The height is the ____ **distance from the base to the top corner.**

La altura es la distancia ____ *desde la base hasta la esquina de arriba.*

It is different from the slanted side because ____ .

Es diferente del lado inclinado porque ____ .

A triangle is half of a ____ **with the same base and height.**

Un triángulo es la mitad de un ____ *con la misma base y altura.*

Try It

Solve on your own. Check the answer key when you are done.

1. A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

- A. 6 ft
- B. 3 ft
- C. 16 ft
- D. 12 ft

Show your work:

2. What is the area of a triangle with base 14 ft and height 6 ft?

- A. 42 sq ft
- B. 84 sq ft
- C. 20 sq ft
- D. 42 ft

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

Sentence starter: The rectangle's area is ____ because _____. Each triangle's area is ____ because a diagonal splits a rectangle into ____ equal triangles.

Show your work:

Reflect — Exit Ticket

A triangle has a base of 11 inches and a height of 8 inches. What is its area?

- A. 44 sq in
- B. 88 sq in
- C. 19 sq in
- D. 44 in

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $6 \text{ ft} - 24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6 \text{ ft}$.
2. **Try It 2:** A. $42 \text{ sq ft} - A = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42 \text{ sq ft}$.
3. **Exit Ticket:** A. $44 \text{ sq in} - A = \frac{1}{2} \times 11 \times 8 = \frac{1}{2} \times 88 = 44 \text{ square inches}$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Area is how much space is inside a flat shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).